

1. Getting Started - Tableau Workspace, Tableau terminologies, basic functionalities.

1) Install & open Tableau

1. Download Tableau Desktop (trial) or Tableau Public from the Tableau website and install it.
 2. Open Tableau. You'll land on the **Start** screen (Connect pane on left, sample data, recent workbooks).
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2) Connect to data (first hands-on)

Use a simple file to practice — e.g., an Excel file with columns: Order Date, Region, Product, Sales, Profit, Quantity.

1. On Start → **Connect** → choose **Microsoft Excel** (or **Text file / CSV**).
 2. Browse and open your file. Tableau shows a data source tab with a preview.
 3. If multiple sheets, drag the sheet you want to the canvas. Optionally switch to the **Data Interpreter** if data is messy.
 4. Click **Sheet 1** at bottom to open a Worksheet.
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3) Understand the Tableau workspace (what you see)

- **Menu Bar / Toolbar** — file, undo, save, export.
 - **Data Pane (left)** — shows your data source, with **Dimensions** (categorical) and **Measures** (numeric).
 - **Columns** and **Rows** shelves — place fields here to shape the chart.
 - **Marks card** — controls Color, Size, Label, Tooltip, Shape, Detail.
 - **Filters shelf** — add global or sheet-level filters.
 - **Pages shelf** — create animations/step-throughs (optional).
 - **Sheet tabs (bottom)** — switch between worksheets, dashboards, stories.
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4) Build your first charts — step-by-step examples

Example A — Bar chart (Sales by Region)

1. From Data Pane, drag **Region** (Dimension) to **Columns**.
2. Drag **Sales** (Measure) to **Rows**.

3. Tableau auto-creates a bar chart.
4. On **Marks** card, click **Label** → check **Show mark labels** to display values.
5. Sort descending: click the sort icon on the Region pill in Columns.

Example B — Line chart (Sales over time)

1. Drag **Order Date** to **Columns**. If it defaults to YEAR, fine — you can change granularity (right-click the pill → **Month, Quarter**, etc.).
2. Drag **Sales** to **Rows**.
3. On Marks → change type to **Line**.
4. Optional: drag **Region** to **Color** to show separate lines per region.

Example C — Scatter (Profit vs Sales)

1. Drag **Sales** to Columns and **Profit** to Rows.
2. Change Marks → **Circle**.
3. Drag **Product** or **Category** to **Detail** to show separate points.
4. Drag **Quantity** to **Size** to scale points by quantity.

Example D — Map (if data has geographic fields)

1. Ensure you have Country, State, or City as Dimensions.
2. Double-click **State** — Tableau builds a map automatically.
3. Drag **Sales** to **Color** to show intensity.

5) Filters, sorting & sets

- **Add a filter:** drag a field (e.g., Category) to Filters shelf → choose values to keep/exclude → click the filter pill → **Show Filter** to expose a control on the worksheet (dropdown, single value list, multi-select).
- **Top N filter:** In Filters → pick a measure → use **Top** tab to show top N by a measure (e.g., Top 10 products by Sales).
- **Sets:** Right-click a Dimension → **Create** → **Set** to define custom grouped sets for analysis.

6) Calculated fields & aggregation examples

Create new fields when you need derived metrics.

1. Right-click in Data Pane → **Create** → **Calculated Field**.
2. Example: **Profit Ratio**
3. Name: Profit Ratio
4. Formula: $\text{SUM}([\text{Profit}]) / \text{SUM}([\text{Sales}])$

Then use it in views (drag to Columns, Color, Tooltip).

5. Common functions: IF / ELSEIF / ELSE, DATEPART(), ZN(), WINDOW_SUM() — use them to transform/aggregate data.

7) Quick table calculations

Right-click a measure on Rows/Columns → **Quick Table Calculation** → choose **Running Total**, **Percent of Total**, **Difference**, etc. Useful for cumulative sales or percent-share views.

8) Build a basic dashboard

1. Click **New Dashboard** (tab at bottom).
2. On left, you see **Sheets** — drag Sheet 1, Sheet 2, etc. into the canvas.
3. Resize, align and use **Tiled** or **Floating** layout.
4. Add interactivity: select a sheet → **Use as Filter** (top-right corner of visualization) so clicking a bar filters the whole dashboard.
5. Add titles, text boxes, images from the Left pane.
6. Preview dashboard at different sizes (Device Preview) for mobile/tablet.

9) Story (optional)

Create a story to walk viewers through steps: **New Story** → drag sheets or dashboards as story points, add captions and navigation.

10) Publish & share

- **Save locally:** File → Save As → .twb or .twbx (packaged workbook includes data).
- **Publish to Tableau Public:** Server → Tableau Public → Sign in → Publish workbook (publicly visible).
- **Publish to Tableau Server / Online:** same menu, requires account.

11) Practical beginner checklist (do these)

1. Open sample dataset (Superstore or your Excel).
2. Create at least: bar, line, scatter, map (if geo data exists), and a pie or donut.
3. Add a filter and expose “Show filter” for user control.
4. Create one calculated field (Profit Ratio) and use it.
5. Build a dashboard with 2–3 sheets and enable **Use as Filter**.
6. Save as .twbx.

12) Helpful tips & best practices

- **Dimensions vs Measures:** Dimensions slice, measures aggregate.
- **Keep data clean:** fix column names and data types at the data source tab.
- **Use Show Me:** highlights chart types suitable for selected fields.
- **Tooltips:** edit to provide context (click Tooltip on Marks card).
- **Color choice:** use Tableau palettes; avoid too many colors.
- **Performance:** limit rows, extract data (Data → Extract) for large datasets.
- **Keyboard shortcuts:** Ctrl+Z (undo), Ctrl+S (save), double-click field to auto-place.

13) Small project to practice (30–60 minutes)

1. Open workbook with sales data.
2. Create a bar chart: Sales by Category (with labels).
3. Create line chart: Monthly Sales for last 12 months.
4. Create map (if available): Sales by State.
5. Create calculated field Profit Ratio.
6. Build dashboard with the three sheets; add a category filter visible on dashboard; enable “Use as Filter” on the bar chart.
7. Save and export dashboard as an image/PDF.

14) Common beginner errors & fixes

- **Blank map** → ensure geographic role assigned (right-click field → Geographic Role → State/Country).
 - **Dates not recognized** → change data type to Date in Data Pane.
 - **Unexpected aggregation (SUM vs AVG)** → click measure pill → Aggregation → change.
 - **Too many marks / slow workbook** → use extracts, filters, or limit marks.
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15) Next resources & learning path

- Tableau official help (you listed these links — great): workspace, sheets, terms, functions.
- Tableau Public gallery: explore real dashboards for ideas.
- Short tutorials: build bar/line/maps, calculated fields, parameters.
- Practice: import your own dataset (CSV/Excel) and repeat steps above.